

PulseGuard — a multi-event intraoperative early-warning system

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Live demo: aastikrajan.github.io/pulseguard Code + full benchmark: github.com/AastikRajan/pulseguard

Research prototype — not clinical software. Every figure below is out-of-sample, on a single locked test split.

PulseGuard watches all of a patient’s vital signs during surgery and warns, minutes ahead, that they are about to deteriorate — saying which of nine problems is coming and why. It was built solo on public data (**VitalDB, 5,151 surgeries** from Seoul National University Hospital). Most of the effort went not into the model itself but into **whether it can be trusted**: calibrating its alarms, testing whether it survives a move to a different hospital, mapping exactly where it fails, and separating real causal effects from confounding.

Data and evaluation

- VitalDB: 5,151 surgeries, 35 signals at 1 Hz (plus 500 Hz waveforms).
- Locked split (seed fixed once): 20% of cases held out = **1,030 test cases / 303,909 time windows**. No patient appears in both training and test.
- Labels come from the recorded vitals and events; nothing below is tuned on the test set.

What the system does

Capability	Result (held-out test set)
Unified early-warning — any deterioration, 5-min horizon	AUPRC 0.454 · AUROC 0.849 · ~4.3-min median lead (base rate 12%)
Alarm precision after conformal calibration	62–71% real vs ~12% for deployed monitors; ~300× fewer alarms
Multi-event detection — 9 distinct events	per-event AUROC 0.74–0.96
Cross-hospital, zero-shot (no retraining)	AUROC 0.78–0.80 Seoul → UC-Irvine
Causal: does low blood pressure cause kidney injury?	RR 1.77 → 1.07 mostly confounding

The nine events: hypoxemia, hypotension, bradycardia, tachycardia, bronchospasm, hypercapnia, ST-deviation (ischemia), light anesthesia (awareness), and apnea. Each is detected by its own model — a single 9-way classifier does worse, because these events often happen together.

How it works, in brief

- Every 30-second window of vitals is turned into features — current levels, short-term trends, variability, cross-signal coupling, and leak-free event-history (what has already happened in this case). A gradient-boosted model scores the chance of deterioration in the next five minutes.
- Raw scores are turned into trustworthy alarms by **conformal calibration** (which gives a precision target), then **episode-grouping** collapses a burst of firings into a single alert — that’s the ~300× reduction.
- For cross-hospital use the model keeps only the channels every monitor has (heart rate, SpO₂); adding rich, source-specific channels helps within one hospital but *hurts* transfer to another.

Causal sub-study — does hypotension cause kidney injury?

A separate arm asks whether intraoperative low blood pressure *causes* postoperative acute kidney injury. The crude association is RR 1.77. After deconfounding for recorded patient factors (g-formula), it falls to **RR 1.07 — about 90% of the apparent effect is confounding by indication** (sicker patients get both). The null replicates on a second hospital and matches the 2025 randomized trials, where preventing hypotension did not prevent kidney injury.

Where it can't be trusted (the boundary I mapped)

- **Intraoperative hypotension is close to unpredictable** — system recall ~0.12. I confirmed this seven independent ways (operating-point analysis, lead-signal tests, critical-slowness, a dynamical network biomarker, short-horizon reachability, phenotype analysis, and 500 Hz heart-rate-variability), so I report it as a hard limit instead of tuning around it.
- Four signals carry almost all the information: heart rate, SpO₂, the pressure line, and the ventilator. Heart rate and SpO₂ have no substitute — so a **low-cost version using only ECG + pulse-oximetry** is plausible where arterial lines aren't available.

Why the numbers are believable

- One locked test split, chosen once and never tuned on.
- I caught a data leak in my own pipeline that inflated the headline (0.48 → 0.454 once fixed) and quarantined it.
- Every null and failure is reported, not buried — the honest boundary above is part of the result.

What's new, and what I'm looking for

I haven't found a published system that combines all of **multi-event early-warning, zero-shot cross-hospital transfer, a causal analysis, and calibrated alarms** — alongside an explicit map of what it cannot predict. The contribution is integration, interpretability, and rigor rather than a new biomarker. **I'm looking for feedback, a research role, or a collaborator** to take the strongest pieces — the cross-hospital transfer and the honest benchmark — toward a paper and external validation.